Estimating and Predicting Player Performance using Hierarchical Statistical Models

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2026-09-01

```
library(readxl)
```

```
## Warning: package 'readxl' was built under R version 4.3.3
```

```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.3.3
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(knitr)
```

```
## Warning: package 'knitr' was built under R version 4.3.3
```

```
library(kableExtra)
```

```
##
## Attaching package: 'kableExtra'
```

```
## The following object is masked from 'package:dplyr':
##
##     group_rows
```

```
excel_path <- "C:/Users/youpindao/OneDrive/Desktop/documents/lol_player_stats.xlsx"
```

Individual Player Match Data

The following tables contain the last 50 Ranked Solo/Duo games for each player.

Solar

```
solar_matches <- read_excel(
  excel_path,
  sheet = "Solar"
)

solar_matches <- solar_matches %>%
  mutate(
    position = ifelse(position == "UTILITY", "SUPPORT", position),
    kda = (kills + assists) / pmax(deaths, 1)
  )

kable(
  solar_matches,
  caption = "Solar — Last 50 Ranked Solo/Duo Games",
  digits = 2,
  row.names = FALSE
) %>%
kable_styling(
  font_size = 8,
  full_width = TRUE
)
```

```
## Warning: 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
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## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```

Solar — Last 50 Ranked Solo/Duo Games

match_id	game_time	duration_min	champion	position	win	kills	deaths	assists	cs	cs_per_min	damage	gold	vision_score	kda
NA1_5632396519	2026-08-31 12:29:45	15.38	JarvanIV	JUNGLE	FALSE	1	5	4	74	4.81	5116	4974	9	1.00
NA1_5632400962	2026-08-31 11:53:34	30.52	JarvanIV	JUNGLE	TRUE	4	9	18	161	5.28	15659	12851	19	2.44
NA1_5632390617	2026-08-31 11:19:34	29.23	MasterYi	JUNGLE	FALSE	7	16	7	107	3.66	24922	11094	11	0.88
NA1_5632385389	2026-08-31 10:44:22	26.77	Nocturne	JUNGLE	TRUE	6	10	9	119	4.45	16950	10543	10	1.50
NA1_5632385123	2026-08-31 10:14:24	23.15	Veigar	MIDDLE	TRUE	1	4	3	166	7.17	7142	8826	10	1.00
NA1_5632379259	2026-08-31 09:27:24	36.68	Nocturne	JUNGLE	TRUE	14	13	21	210	5.72	48887	18517	30	2.69
NA1_5632369454	2026-08-31 09:07:52	15.32	MasterYi	JUNGLE	FALSE	3	8	1	69	4.50	9417	5301	8	0.50
NA1_5632366150	2026-08-31 08:34:07	29.63	MasterYi	MIDDLE	FALSE	3	17	4	130	4.39	24098	8752	12	0.41
NA1_5632359011	2026-08-31 07:50:25	39.55	JarvanIV	JUNGLE	FALSE	3	15	15	188	4.75	18681	13349	29	1.20
NA1_5632339384	2026-08-31 07:14:19	29.15	MasterYi	JUNGLE	FALSE	3	13	5	116	3.98	16377	9752	16	0.62
NA1_5632321324	2026-08-31 06:33:37	33.12	Zac	JUNGLE	FALSE	2	14	8	139	4.20	11595	9458	26	0.71
NA1_5632314277	2026-08-31 06:10:46	15.55	MasterYi	JUNGLE	FALSE	3	5	2	83	5.34	8720	5859	8	1.00
NA1_5632293880	2026-08-31 05:31:32	34.70	MasterYi	JUNGLE	FALSE	8	19	3	162	4.67	21176	12793	27	0.58
NA1_5632268108	2026-08-31 04:55:55	26.75	MasterYi	JUNGLE	TRUE	15	5	12	198	7.40	30925	15838	20	5.40
NA1_5632213898	2026-08-31 03:44:23	35.92	JarvanIV	JUNGLE	TRUE	3	10	19	176	4.90	17259	13053	21	2.20
NA1_5632185152	2026-08-31 03:23:16	16.08	Lux	SUPPORT	FALSE	2	9	2	15	0.93	7078	4803	18	0.44
NA1_5632147615	2026-08-31 02:48:24	27.03	MasterYi	JUNGLE	FALSE	0	9	4	147	5.44	14967	8127	23	0.44
NA1_5632038419	2026-08-31 00:58:29	33.23	MasterYi	JUNGLE	FALSE	4	15	7	132	3.97	26992	10928	23	0.73
NA1_5632001602	2026-08-31 00:17:46	21.65	MasterYi	JUNGLE	FALSE	4	10	2	111	5.13	11107	8660	12	0.60
NA1_5631971216	2026-08-30 23:36:16	31.13	Lux	SUPPORT	FALSE	0	9	10	46	1.48	21071	8199	55	1.11
NA1_5631945761	2026-08-30 23:02:13	29.32	MasterYi	JUNGLE	FALSE	4	8	3	153	5.22	25228	9316	22	0.88
NA1_5631585464	2026-08-30 09:35:43	26.95	MasterYi	JUNGLE	FALSE	6	16	13	90	3.34	19438	9318	20	1.19
NA1_5631584844	2026-08-30 08:57:34	30.55	MasterYi	JUNGLE	FALSE	3	15	3	155	5.07	22010	10727	25	0.40

match_id	game_time	duration_min	champion	position	win	kills	deaths	assists	cs	cs_per_min	damage	gold	vision_score	kda
NA1_5631580790	2026-08-30 08:26:41	23.50	MasterYi	JUNGLE	TRUE	10	10	2	130	5.53	22384	10541	10	1.20
NA1_5631560543	2026-08-30 08:03:10	17.58	MasterYi	JUNGLE	TRUE	4	5	3	102	5.80	7305	6836	16	1.40
NA1_5631556415	2026-08-30 07:53:59	1.68	MasterYi	JUNGLE	FALSE	0	0	0	4	2.38	0	655	0	0.00
NA1_5631443265	2026-08-30 04:53:38	32.45	MasterYi	JUNGLE	TRUE	18	13	11	146	4.50	47938	16191	39	2.23
NA1_5631391763	2026-08-30 03:52:48	40.48	MasterYi	JUNGLE	TRUE	15	15	13	223	5.51	48132	19112	49	1.87
NA1_5631331712	2026-08-30 03:05:33	27.63	MasterYi	JUNGLE	TRUE	18	13	8	122	4.41	31839	13653	28	2.00
NA1_5631291834	2026-08-30 02:08:14	45.75	MasterYi	JUNGLE	TRUE	25	13	11	258	5.64	67297	25482	48	2.77
NA1_5631252396	2026-08-30 01:33:44	26.05	MasterYi	JUNGLE	FALSE	10	13	7	112	4.30	22363	10123	23	1.31
NA1_5631206371	2026-08-30 00:36:45	24.72	Jax	JUNGLE	FALSE	9	10	2	90	3.64	13964	9682	10	1.10
NA1_5631149786	2026-08-29 23:11:20	23.18	JarvanIV	JUNGLE	TRUE	6	4	6	147	6.34	8606	10024	15	3.00
NA1_5631101208	2026-08-29 22:01:24	38.12	MasterYi	JUNGLE	TRUE	6	14	19	210	5.51	26118	15940	29	1.79
NA1_5631082017	2026-08-29 21:28:03	23.25	JarvanIV	JUNGLE	TRUE	2	6	15	155	6.67	10892	9911	18	2.83
NA1_5631046569	2026-08-29 20:49:30	21.98	MasterYi	JUNGLE	TRUE	14	2	9	181	8.23	22806	12538	20	11.50
NA1_5631024505	2026-08-29 20:05:14	36.73	MasterYi	JUNGLE	TRUE	10	9	11	238	6.48	31961	16937	29	2.33
NA1_5630809645	2026-08-29 09:20:15	28.40	MasterYi	JUNGLE	TRUE	11	10	10	139	4.89	25816	12216	24	2.10
NA1_5630750172	2026-08-29 06:33:24	27.47	MasterYi	MIDDLE	TRUE	16	6	3	160	5.83	30974	14678	18	3.17
NA1_5630726933	2026-08-29 05:51:58	34.27	MasterYi	JUNGLE	TRUE	10	8	8	239	6.97	24890	16087	38	2.25
NA1_5630692696	2026-08-29 05:21:56	24.85	MasterYi	JUNGLE	TRUE	21	6	3	150	6.04	34692	14189	28	4.00
NA1_5630663770	2026-08-29 04:33:56	41.13	MasterYi	JUNGLE	TRUE	19	15	12	245	5.96	54355	19969	36	2.07
NA1_5630615696	2026-08-29 03:57:17	29.40	Veigar	SUPPORT	TRUE	2	13	5	44	1.50	17917	8931	49	0.54
NA1_5630395918	2026-08-28 23:25:40	25.95	MasterYi	JUNGLE	FALSE	5	9	1	145	5.59	13329	9728	24	0.67
NA1_5629337346	2026-08-27 06:35:05	21.67	Veigar	SUPPORT	FALSE	4	7	8	36	1.66	13272	7599	21	1.71
NA1_5629320664	2026-08-27 06:12:41	15.75	MasterYi	JUNGLE	FALSE	2	8	1	86	5.46	7087	5432	8	0.38
NA1_5628679317	2026-08-26 07:52:06	36.88	MasterYi	JUNGLE	TRUE	5	5	6	206	5.59	25308	13404	26	2.20
NA1_5628660663	2026-08-26 07:08:16	31.23	MasterYi	JUNGLE	TRUE	23	10	7	224	7.17	49224	19776	31	3.00
NA1_5628108174	2026-08-25 07:46:15	33.22	MasterYi	JUNGLE	TRUE	13	8	9	204	6.14	37843	15768	28	2.75
NA1_5628084345	2026-08-25 06:41:48	29.87	MasterYi	JUNGLE	FALSE	1	16	8	121	4.05	21466	9234	20	0.56

Faker

```
faker_matches <- read_excel(
  excel_path,
  sheet = "Faker"
)

faker_matches <- faker_matches %>%
  mutate(
    position = ifelse(position == "UTILITY", "SUPPORT", position),
    kda = (kills + assists) / pmax(deaths, 1)
  )

kable(
  faker_matches,
  caption = "Faker — Last 50 Ranked Solo/Duo Games",
  digits = 2,
  row.names = FALSE
) %>%
kable_styling(
  font_size = 8,
  full_width = TRUE
)
```

```
## Warning: 'xfun::attr()' is deprecated.
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```

Faker — Last 50 Ranked Solo/Duo Games

match_id	game_time	duration_min	champion	position	win	kills	deaths	assists	cs	cs_per_min	damage	gold	vision_score	kda
KR_8363832456	2026-08-31 17:29:06	18.63	Vi	JUNGLE	TRUE	7	1	5	146	7.84	8595	9302	23	12.00
KR_8363818231	2026-08-31 17:01:18	19.57	Yone	MIDDLE	TRUE	13	0	4	208	10.63	31435	12645	21	17.00
KR_8363780662	2026-08-31 16:26:47	26.70	Ryze	MIDDLE	FALSE	5	5	5	239	8.95	23265	11482	34	2.00
KR_8363741424	2026-08-31 15:45:41	29.13	Chogath	JUNGLE	FALSE	4	9	8	172	5.90	17766	10217	44	1.33
KR_8362592055	2026-08-30 17:56:15	21.38	Yone	MIDDLE	TRUE	6	2	7	223	10.43	22041	11848	34	6.50
KR_8362556542	2026-08-30 17:17:48	26.18	Anivia	MIDDLE	FALSE	4	4	4	261	9.97	18202	11253	44	2.00
KR_8362528546	2026-08-30 16:50:03	21.23	Nocturne	JUNGLE	FALSE	4	6	2	109	5.13	9986	7569	25	1.00
KR_8362507352	2026-08-30 16:25:25	15.18	Galio	MIDDLE	TRUE	1	1	8	121	7.97	6049	5696	20	9.00
KR_8362446827	2026-08-30 15:42:33	29.68	Cassiopeia	MIDDLE	TRUE	7	0	10	291	9.80	35132	14777	46	17.00
KR_8361620721	2026-08-30 08:36:15	25.95	Yone	MIDDLE	FALSE	2	11	6	168	6.47	16899	9113	27	0.73
KR_8359150346	2026-08-28 18:10:15	19.62	Locke	MIDDLE	TRUE	6	2	5	150	7.65	14881	8772	18	5.50
KR_8359106011	2026-08-28 17:21:16	25.28	TwistedFate	MIDDLE	FALSE	3	3	7	244	9.65	13079	12805	51	3.33
KR_8359040829	2026-08-28 16:41:36	28.97	Locke	MIDDLE	TRUE	13	5	9	242	8.35	24853	15752	33	4.40
KR_8358985418	2026-08-28 16:05:28	18.32	Jax	TOP	TRUE	5	3	4	114	6.22	13850	7137	14	3.00
KR_8358612825	2026-08-28 13:19:44	46.53	Shen	SUPPORT	FALSE	0	12	17	50	1.07	10338	12636	192	1.42
KR_8358130638	2026-08-28 08:08:04	37.57	Cassiopeia	MIDDLE	FALSE	6	7	14	293	7.80	39669	16080	55	2.86
KR_8357623928	2026-08-27 17:41:52	15.55	Yone	MIDDLE	TRUE	3	2	4	130	8.36	11232	6361	9	3.50
KR_8357592878	2026-08-27 16:57:44	27.77	KSante	TOP	FALSE	0	9	6	207	7.45	12168	8989	27	0.67
KR_8357547284	2026-08-27 16:04:30	29.92	Orianna	MIDDLE	FALSE	2	4	10	267	8.92	23448	12120	24	3.00
KR_8357486357	2026-08-27 15:21:13	29.95	Akali	MIDDLE	TRUE	10	6	7	253	8.45	31097	14830	28	2.83
KR_8356356798	2026-08-26 17:08:52	20.07	Galio	MIDDLE	TRUE	5	1	11	168	8.37	15625	9621	16	16.00
KR_8356353386	2026-08-26 16:44:37	18.37	Cassiopeia	MIDDLE	TRUE	1	3	2	165	8.98	12779	6923	18	1.00
KR_8356299807	2026-08-26 16:07:41	30.40	Vi	JUNGLE	FALSE	9	8	6	180	5.92	22078	13391	34	1.88
KR_8356258475	2026-08-26 15:27:58	18.38	Leblanc	MIDDLE	TRUE	10	0	4	177	9.63	20537	9648	19	14.00
KR_8356197368	2026-08-26 14:45:46	24.10	Yone	MIDDLE	FALSE	11	9	5	203	8.42	30941	14230	24	1.78
KR_8351445185	2026-08-22 17:31:36	24.67	Gnar	TOP	FALSE	2	8	2	197	7.99	15353	8792	33	0.50
KR_8351420264	2026-08-22 17:11:04	15.37	Cassiopeia	MIDDLE	FALSE	3	4	2	105	6.83	8630	5518	15	1.25
KR_8351358536	2026-08-22 16:29:46	36.60	Gwen	TOP	TRUE	1	3	6	326	8.91	20575	15142	32	2.33
KR_8351290867	2026-08-22 15:53:28	28.87	Ambessa	MIDDLE	FALSE	1	8	7	216	7.48	18201	9722	43	1.00
KR_8348580255	2026-08-20 17:21:53	37.38	Akali	MIDDLE	FALSE	11	11	7	295	7.89	40933	17640	44	1.64
KR_8348556318	2026-08-20 16:44:32	30.30	Locke	MIDDLE	TRUE	19	9	10	175	5.78	48426	16076	29	3.22
KR_8348484328	2026-08-20 15:38:36	34.38	Lucian	BOTTOM	TRUE	9	11	15	251	7.30	37730	16406	63	2.18
KR_8348437365	2026-08-20 14:59:44	29.30	Nasus	TOP	TRUE	6	6	8	185	6.31	24570	11391	21	2.33
KR_8347487807	2026-08-19 17:19:25	28.92	Cassiopeia	MIDDLE	FALSE	2	4	9	258	8.92	25909	11390	42	2.75
KR_8347417538	2026-08-19 15:48:55	21.22	Nasus	MIDDLE	FALSE	2	6	0	190	8.96	14046	8232	17	0.33
KR_8347379120	2026-08-19 15:12:02	31.15	Cassiopeia	MIDDLE	TRUE	21	2	9	290	9.31	48944	18858	51	15.00
KR_8346704580	2026-08-19 06:27:39	29.50	Viktor	MIDDLE	FALSE	3	5	3	299	10.14	24121	12163	37	1.20
KR_8346669878	2026-08-19 05:47:34	32.42	Nasus	TOP	TRUE	3	8	4	219	6.76	24272	11869	39	0.88

match_id	game_time	duration_min	champion	position	win	kills	deaths	assists	cs	cs_per_min	damage	gold	vision_score	kda
KR_8346650523	2026-08-19 05:08:00	30.10	Nasus	MIDDLE	FALSE	3	12	12	182	6.05	33356	10444	23	1.25
KR_8346635676	2026-08-19 04:41:35	21.38	Cassiopeia	MIDDLE	TRUE	15	1	0	218	10.19	22371	11707	21	15.00
KR_8346614705	2026-08-19 04:07:06	25.45	Cassiopeia	MIDDLE	TRUE	9	3	12	252	9.90	32842	13944	16	7.00
KR_8346334525	2026-08-18 16:08:02	24.80	Nasus	TOP	FALSE	4	8	4	134	5.40	18071	9043	24	1.00
KR_8346280152	2026-08-18 15:18:36	42.97	Anivia	MIDDLE	TRUE	8	5	19	260	6.05	47511	15832	53	5.40
KR_8346231786	2026-08-18 14:50:36	15.32	Syndra	MIDDLE	TRUE	3	0	0	167	10.90	8172	7586	13	3.00
KR_8346212209	2026-08-18 14:31:39	15.33	Yorick	TOP	TRUE	1	1	6	140	9.13	11111	6806	15	7.00
KR_8344927004	2026-08-17 13:17:10	31.52	Cassiopeia	MIDDLE	FALSE	14	6	10	243	7.71	43681	15587	36	4.00
KR_8343739718	2026-08-16 18:02:49	33.07	Syndra	MIDDLE	TRUE	4	7	15	258	7.80	30037	13416	32	2.71
KR_8342112007	2026-08-15 18:00:43	29.93	Jax	TOP	FALSE	5	1	6	247	8.25	19300	12122	44	11.00
KR_8342058083	2026-08-15 17:12:57	15.43	Locke	MIDDLE	TRUE	5	1	3	140	9.07	10980	7434	9	8.00
KR_8341992710	2026-08-15 16:30:26	27.38	Orianna	MIDDLE	FALSE	2	2	2	286	10.44	24217	11374	36	2.00

Chovy

```
chovy_matches <- read_excel(
  excel_path,
  sheet = "Chovy"
)

chovy_matches <- chovy_matches %>%
  mutate(
    position = ifelse(position == "UTILITY", "SUPPORT", position),
    kda = (kills + assists) / pmax(deaths, 1)
  )

kable(
  chovy_matches,
  caption = "Chovy — Last 50 Ranked Solo/Duo Games",
  digits = 2,
  row.names = FALSE
) %>%
kable_styling(
  font_size = 8,
  full_width = TRUE
)
```

```
## Warning: 'xfun::attr()' is deprecated.
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## See help("Deprecated")
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## See help("Deprecated")
```

Chovy — Last 50 Ranked Solo/Duo Games

match_id	game_time	duration_min	champion	position	win	kills	deaths	assists	cs	cs_per_min	damage	gold	vision_score	kda
KR_8363812949	2026-08-31 17:11:58	22.72	Syndra	MIDDLE	FALSE	2	4	4	223	9.82	17049	9551	27	1.50
KR_8363783316	2026-08-31 16:25:59	36.05	Shen	SUPPORT	FALSE	2	12	21	24	0.67	15300	9894	129	1.92
KR_8363756414	2026-08-31 15:55:01	23.82	Ahri	MIDDLE	FALSE	7	3	7	260	10.92	21159	12630	45	4.67
KR_8363698290	2026-08-31 15:09:07	15.33	Cassiopeia	MIDDLE	TRUE	3	1	2	156	10.17	8642	6852	17	5.00
KR_8363636556	2026-08-31 14:32:20	23.07	Syndra	MIDDLE	FALSE	7	6	2	201	8.71	27414	10882	22	1.50
KR_8362579482	2026-08-30 17:52:26	31.35	Kaisa	BOTTOM	TRUE	6	7	11	267	8.52	27159	15019	42	2.43
KR_8362545352	2026-08-30 17:06:17	35.48	Caitlyn	BOTTOM	TRUE	14	3	5	421	11.86	74605	22986	82	6.33

match_id	game_time	duration_min	champion	position	win	kills	deaths	assists	cs	cs_per_min	damage	gold	vision_score	kda
KR_8362509818	2026-08-30 16:33:41	25.58	Vex	MIDDLE	FALSE	5	5	4	256	10.01	24745	12252	38	1.80
KR_8362493327	2026-08-30 16:10:37	18.63	Cassiopeia	MIDDLE	TRUE	4	2	2	182	9.77	16386	7912	32	3.00
KR_8362440181	2026-08-30 15:27:50	27.73	Ryze	MIDDLE	TRUE	8	4	13	297	10.71	35571	14055	33	5.25
KR_8360812743	2026-08-29 17:19:42	25.50	Anivia	MIDDLE	FALSE	1	4	1	242	9.49	18503	9053	35	0.50
KR_8360724805	2026-08-29 16:22:03	40.62	Akali	MIDDLE	TRUE	8	6	9	328	8.08	42628	16804	47	2.83
KR_8360652778	2026-08-29 15:42:46	22.37	TwistedFate	MIDDLE	FALSE	4	4	6	243	10.86	15055	12259	33	2.50
KR_8360572555	2026-08-29 15:02:26	31.55	Syndra	MIDDLE	TRUE	6	7	9	345	10.94	40087	16615	50	2.14
KR_8360522108	2026-08-29 14:38:02	18.58	Cassiopeia	MIDDLE	TRUE	5	3	8	171	9.20	18325	8709	16	4.33
KR_8360417452	2026-08-29 13:56:37	27.18	Shen	SUPPORT	TRUE	5	3	16	62	2.28	11858	9746	120	7.00
KR_8358997067	2026-08-28 16:05:49	41.98	Caitlyn	BOTTOM	TRUE	5	2	14	496	11.81	53133	23396	73	9.50
KR_8358925560	2026-08-28 15:35:47	23.42	Jhin	BOTTOM	FALSE	5	6	4	206	8.80	18042	12085	35	1.50
KR_8358842533	2026-08-28 14:55:32	23.28	Locke	MIDDLE	FALSE	2	5	2	211	9.06	13040	8789	26	0.80
KR_8358783339	2026-08-28 14:29:13	21.10	Anivia	MIDDLE	FALSE	1	7	5	168	7.96	17679	7570	21	0.86
KR_8358453026	2026-08-28 11:55:40	27.33	Shen	SUPPORT	FALSE	6	8	14	36	1.32	14065	9081	106	2.50
KR_8358353922	2026-08-28 11:05:30	24.02	Olaf	TOP	TRUE	7	5	8	236	9.83	24172	12830	23	3.00
KR_8358296494	2026-08-28 10:19:03	33.05	Orianna	MIDDLE	FALSE	11	3	9	327	9.89	39511	16108	42	6.67
KR_8358227112	2026-08-28 09:21:47	33.60	Viktor	MIDDLE	FALSE	8	5	13	302	8.99	61897	15640	54	4.20
KR_8357588358	2026-08-27 16:43:37	15.72	TwistedFate	MIDDLE	TRUE	2	1	4	169	10.75	8806	7947	14	6.00
KR_8357547284	2026-08-27 16:04:30	29.92	Locke	MIDDLE	TRUE	4	5	4	229	7.65	15409	11527	41	1.60
KR_8357479987	2026-08-27 15:23:33	30.13	Ezreal	BOTTOM	TRUE	2	3	5	333	11.05	23322	14611	58	2.33
KR_8357424665	2026-08-27 14:47:03	25.88	Cassiopeia	MIDDLE	TRUE	12	3	2	266	10.28	31621	15584	35	4.67
KR_8357359474	2026-08-27 14:09:14	29.52	Shen	SUPPORT	FALSE	4	10	15	53	1.80	15176	10212	114	1.90
KR_8349804117	2026-08-21 16:46:53	15.88	Galio	MIDDLE	TRUE	4	2	8	135	8.50	10790	6706	22	6.00
KR_8349758521	2026-08-21 16:20:56	21.03	Viktor	MIDDLE	TRUE	4	2	2	230	10.94	23557	9980	24	3.00
KR_8349717605	2026-08-21 15:57:24	19.25	Cassiopeia	MIDDLE	TRUE	4	0	5	190	9.87	11390	8066	34	9.00
KR_8349642919	2026-08-21 15:21:23	26.98	Ryze	MIDDLE	TRUE	7	6	11	307	11.38	25724	14749	35	3.00
KR_8348250207	2026-08-20 13:08:26	34.10	Ryze	MIDDLE	TRUE	7	2	9	374	10.97	29861	17315	55	8.00
KR_8346352992	2026-08-18 16:38:42	22.72	Lissandra	MIDDLE	FALSE	2	3	3	206	9.07	18623	8775	31	1.67
KR_8346334525	2026-08-18 16:08:02	24.80	Vi	JUNGLE	FALSE	8	5	8	193	7.78	15536	12251	51	3.20
KR_8346280152	2026-08-18 15:18:36	42.97	Cassiopeia	MIDDLE	FALSE	6	8	6	362	8.43	51884	17883	60	1.50
KR_8345228883	2026-08-17 16:43:40	25.08	Galio	MIDDLE	TRUE	10	5	11	218	8.69	26290	12785	27	4.20
KR_8345197591	2026-08-17 16:09:45	27.98	Cassiopeia	MIDDLE	FALSE	7	8	4	254	9.08	30507	13492	43	1.38
KR_8345157429	2026-08-17 15:33:42	15.27	Tristana	MIDDLE	TRUE	1	1	2	163	10.68	12225	6893	17	3.00
KR_8345105327	2026-08-17 14:54:20	32.83	Galio	MIDDLE	FALSE	8	5	11	338	10.29	31914	16158	36	3.80
KR_8345048469	2026-08-17 14:22:57	15.43	Syndra	MIDDLE	FALSE	1	2	2	139	9.01	11058	5808	11	1.50
KR_8344988854	2026-08-17 13:55:57	21.93	Gnar	TOP	FALSE	6	4	3	208	9.48	20402	10762	37	2.25
KR_8344911590	2026-08-17 13:15:08	26.95	Syndra	MIDDLE	FALSE	6	7	8	274	10.17	23944	12826	35	2.00
KR_8341987008	2026-08-15 16:18:56	26.68	Syndra	MIDDLE	TRUE	11	2	8	311	11.66	35150	15926	26	9.50
KR_8341932677	2026-08-15 15:53:04	15.18	Orianna	MIDDLE	TRUE	14	1	8	143	9.42	20473	9938	13	22.00
KR_8341890175	2026-08-15 15:25:26	21.87	Galio	MIDDLE	TRUE	3	1	5	227	10.38	12599	9317	19	8.00
KR_8340960624	2026-08-15 06:12:41	22.58	Akali	MIDDLE	TRUE	12	2	5	207	9.17	24494	11699	21	8.50
KR_8340936163	2026-08-15 05:47:35	15.18	Syndra	MIDDLE	TRUE	2	0	2	157	10.34	10879	7271	15	4.00
KR_8340475076	2026-08-14 16:49:17	18.55	Locke	MIDDLE	TRUE	17	1	4	190	10.24	17841	12320	28	21.00

ShowMaker

```
showmaker_matches <- read_excel(
  excel_path,
  sheet = "ShowMaker"
)

showmaker_matches <- showmaker_matches %>%
  mutate(
    position = ifelse(position == "UTILITY", "SUPPORT", position),
    kda = (kills + assists) / pmax(deaths, 1)
  )

kable(
  showmaker_matches,
  caption = "ShowMaker — Last 50 Ranked Solo/Duo Games",
  digits = 2,
  row.names = FALSE
) %>%
kable_styling(
  font_size = 8,
  full_width = TRUE
)
```

```
## Warning: 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
## Warning: 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```

ShowMaker — Last 50 Ranked Solo/Duo Games

match_id	game_time	duration_min	champion	position	win	kills	deaths	assists	cs	cs_per_min	damage	gold	vision_score	kda
KR_8363837959	2026-08-31 17:57:36	20.68	Sylas	MIDDLE	TRUE	12	0	7	207	10.01	26960	12444	21	19.00
KR_8363824812	2026-08-31 17:32:38	20.00	Ezreal	BOTTOM	TRUE	4	5	7	144	7.20	19442	9782	13	2.20
KR_8363785678	2026-08-31 16:35:18	21.60	Nidalee	TOP	FALSE	7	5	3	163	7.55	24756	9624	12	2.00
KR_8363755873	2026-08-31 16:03:29	22.75	Leblanc	MIDDLE	FALSE	0	4	2	199	8.75	13320	7968	15	0.50
KR_8363381278	2026-08-31 12:24:41	33.85	Jayce	TOP	FALSE	7	11	11	250	7.39	39247	15047	32	1.64
KR_8363299532	2026-08-31 11:44:41	28.78	Nasus	MIDDLE	TRUE	5	5	11	185	6.43	23299	12069	16	3.20
KR_8362579482	2026-08-30 17:52:26	31.35	Nasus	MIDDLE	FALSE	4	10	8	216	6.89	22235	13072	32	1.20
KR_8362545352	2026-08-30 17:06:17	35.48	Ryze	MIDDLE	FALSE	10	4	8	319	8.99	39986	17731	33	4.50
KR_8362455356	2026-08-30 15:39:57	30.65	Malphite	TOP	TRUE	7	5	13	266	8.68	29461	15332	33	4.00
KR_8360489647	2026-08-29 14:28:21	30.30	Chogath	JUNGLE	FALSE	6	3	8	215	7.10	22051	12555	33	4.67
KR_8359142410	2026-08-28 18:05:20	37.35	Akali	MIDDLE	TRUE	7	8	12	252	6.75	32064	14474	29	2.38
KR_8359111611	2026-08-28 17:34:51	25.05	Jayce	MIDDLE	FALSE	2	5	1	237	9.46	18570	9581	23	0.60
KR_8359074607	2026-08-28 17:02:30	22.25	Galio	MIDDLE	TRUE	7	3	10	192	8.63	18114	9940	20	5.67
KR_8358925560	2026-08-28 15:35:47	23.42	Kaisa	BOTTOM	TRUE	8	3	9	205	8.75	22674	12318	22	5.67
KR_8357630325	2026-08-27 17:32:12	27.37	Fizz	MIDDLE	FALSE	8	5	4	178	6.50	14919	12503	23	2.40
KR_8357593280	2026-08-27 16:44:32	40.15	Diana	MIDDLE	FALSE	1	11	6	273	6.80	22147	12593	60	0.64
KR_8356354863	2026-08-26 16:55:22	22.42	Nasus	MIDDLE	TRUE	4	6	10	156	6.96	21496	10121	19	2.33
KR_8356303620	2026-08-26 16:04:53	22.07	Zoe	MIDDLE	TRUE	3	2	5	199	9.02	15622	9318	26	4.00
KR_8355188641	2026-08-25 17:29:06	23.37	Ahri	MIDDLE	TRUE	16	2	6	204	8.73	38819	13510	21	11.00
KR_8355179374	2026-08-25 17:06:24	15.52	Cassiopeia	MIDDLE	TRUE	8	1	4	154	9.92	13683	8337	13	12.00
KR_8354119704	2026-08-24 18:18:17	29.28	Yorick	TOP	FALSE	3	4	3	241	8.23	20390	13994	32	1.50
KR_8349658884	2026-08-21 15:29:03	32.08	Akali	MIDDLE	TRUE	6	9	12	233	7.26	34060	13178	21	2.00

match_id	game_time	duration_min	champion	position	win	kills	deaths	assists	cs	cs_per_min	damage	gold	vision_score	kda
KR_8348539655	2026-08-20 16:33:58	27.38	Akali	MIDDLE	FALSE	8	4	1	217	7.92	31970	11573	17	2.25
KR_8347432922	2026-08-19 16:14:13	29.65	Locke	MIDDLE	FALSE	13	5	5	277	9.34	28211	15666	37	3.60
KR_8346404834	2026-08-18 17:49:36	23.95	Locke	MIDDLE	FALSE	10	6	1	174	7.27	17968	11409	19	1.83
KR_8346388637	2026-08-18 17:15:10	30.58	Locke	SUPPORT	FALSE	10	5	10	51	1.67	20797	12878	117	4.00
KR_8346355706	2026-08-18 16:37:42	16.63	Nasus	MIDDLE	TRUE	3	6	2	109	6.55	10495	6002	9	0.83
KR_8346320115	2026-08-18 15:50:55	32.65	Nasus	MIDDLE	TRUE	10	6	8	233	7.14	28469	13998	49	3.00
KR_8346263505	2026-08-18 15:09:37	25.25	Locke	MIDDLE	FALSE	13	5	3	201	7.96	24215	14178	17	3.20
KR_8345216155	2026-08-17 16:17:12	22.42	Akali	MIDDLE	TRUE	11	2	7	189	8.43	25614	11993	17	9.00
KR_8345187217	2026-08-17 15:52:14	15.23	Karma	SUPPORT	FALSE	0	7	1	25	1.64	5836	3729	21	0.14
KR_8345143330	2026-08-17 15:19:43	25.07	Akali	MIDDLE	TRUE	7	2	6	248	9.89	25005	12608	24	6.50
KR_8343726608	2026-08-16 17:43:05	27.42	Galio	MIDDLE	TRUE	5	2	15	236	8.61	20811	11538	20	10.00
KR_8343690585	2026-08-16 17:11:46	26.37	Locke	MIDDLE	FALSE	7	3	7	207	7.85	19572	11338	22	4.67
KR_8343657367	2026-08-16 16:44:36	22.55	Akali	MIDDLE	FALSE	4	5	5	186	8.25	23949	9737	17	1.80
KR_8343611129	2026-08-16 16:10:56	29.17	Ryze	MIDDLE	TRUE	14	6	11	269	9.22	39972	16040	29	4.17
KR_8339143551	2026-08-13 16:24:33	25.12	Orianna	MIDDLE	TRUE	8	6	9	226	9.00	39093	13084	28	2.83
KR_8339112134	2026-08-13 15:50:31	26.90	Syndra	MIDDLE	FALSE	6	6	5	223	8.29	29812	11466	18	1.83
KR_8336830875	2026-08-11 17:25:09	31.17	Jayce	MIDDLE	FALSE	3	6	6	286	9.18	27505	13124	39	1.50
KR_8336803936	2026-08-11 16:55:12	18.45	Leona	SUPPORT	FALSE	1	4	11	18	0.98	6234	5698	44	3.00
KR_8336794033	2026-08-11 16:26:15	18.30	Syndra	MIDDLE	FALSE	1	4	1	173	9.45	12902	6969	13	0.50
KR_8335681394	2026-08-10 18:34:12	31.63	Taliyah	MIDDLE	FALSE	8	5	13	278	8.79	40286	15554	40	4.20
KR_8335671067	2026-08-10 18:00:56	20.22	Yasuo	MIDDLE	FALSE	3	5	0	172	8.51	10747	8052	25	0.60
KR_8335621991	2026-08-10 17:12:37	28.42	Cassiopeia	MIDDLE	FALSE	7	4	6	235	8.27	29304	12237	19	3.25
KR_8332849943	2026-08-08 17:08:25	16.33	Cassiopeia	MIDDLE	TRUE	2	2	3	131	8.02	12465	6150	11	2.50
KR_8332806352	2026-08-08 16:27:54	19.27	Vi	JUNGLE	TRUE	3	1	8	155	8.04	7493	8781	20	11.00
KR_8332761377	2026-08-08 15:59:58	16.23	Galio	MIDDLE	FALSE	2	4	0	124	7.64	7085	6054	9	0.50
KR_8332703787	2026-08-08 15:32:56	22.90	Galio	MIDDLE	TRUE	9	2	10	191	8.34	27530	11598	20	9.50
KR_8329970260	2026-08-06 16:53:00	26.95	Orianna	MIDDLE	TRUE	8	2	8	265	9.83	30278	15764	35	8.00
KR_8329928094	2026-08-06 16:09:07	21.37	Xerath	MIDDLE	TRUE	2	4	8	177	8.28	17065	8440	14	2.50

Zeka

```
zeka_matches <- read_excel(
  excel_path,
  sheet = "Zeka"
)

zeka_matches <- zeka_matches %>%
  mutate(
    position = ifelse(position == "UTILITY", "SUPPORT", position),
    kda = (kills + assists) / pmax(deaths, 1)
  )

kable(
  zeka_matches,
  caption = "Zeka — Last 50 Ranked Solo/Duo Games",
  digits = 2,
  row.names = FALSE
) %>%
kable_styling(
  font_size = 8,
  full_width = TRUE
)
```

```
## Warning: 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
## Warning: 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```

Zeka — Last 50 Ranked Solo/Duo Games

match_id	game_time	duration_min	champion	position	win	kills	deaths	assists	cs	cs_per_min	damage	gold	vision_score	kda
KR_8363812949	2026-08-31 17:11:58	22.72	Ryze	MIDDLE	TRUE	11	2	6	222	9.77	19427	12169	20	8.50
KR_8363785678	2026-08-31 16:35:18	21.60	Vayne	TOP	TRUE	3	8	3	139	6.44	12534	8125	15	0.75
KR_8363739732	2026-08-31 15:48:06	36.65	Taliyah	MIDDLE	FALSE	9	7	6	306	8.35	32584	16138	55	2.14
KR_8363716357	2026-08-31 15:21:20	15.55	Zeri	MIDDLE	TRUE	6	1	4	176	11.32	14630	9479	22	10.00
KR_8363684150	2026-08-31 14:59:53	15.58	Orianna	MIDDLE	TRUE	4	0	7	157	10.07	17475	7967	16	11.00
KR_8362528546	2026-08-30 16:50:03	21.23	Ziggs	MIDDLE	FALSE	2	8	4	191	9.00	20312	8527	11	0.75
KR_8362490391	2026-08-30 16:09:18	18.77	Leblanc	SUPPORT	TRUE	6	5	10	25	1.33	12775	7320	52	3.20
KR_8362458715	2026-08-30 15:49:47	15.18	Locke	MIDDLE	TRUE	12	2	5	96	6.32	15972	8440	17	8.50
KR_8362061173	2026-08-30 12:25:09	28.20	Corki	BOTTOM	FALSE	2	7	5	247	8.76	22076	11741	25	1.00
KR_8361951802	2026-08-30 11:42:03	16.02	Yone	MIDDLE	TRUE	5	3	8	141	8.80	12819	8174	21	4.33
KR_8361888112	2026-08-30 11:04:23	23.77	Ryze	MIDDLE	FALSE	3	9	8	178	7.49	23421	9477	21	1.22
KR_8361771999	2026-08-30 10:08:19	21.23	Syndra	MIDDLE	FALSE	1	7	3	193	9.09	10811	8001	29	0.57
KR_8360670959	2026-08-29 15:53:50	26.82	Camille	SUPPORT	TRUE	7	9	18	31	1.16	24012	10882	107	2.78
KR_8360591013	2026-08-29 15:06:59	41.83	Jayce	TOP	FALSE	12	11	14	286	6.84	55087	18773	53	2.36
KR_8359101455	2026-08-28 17:22:23	36.98	Hwei	MIDDLE	TRUE	12	7	12	318	8.60	46612	19170	35	3.43
KR_8358997067	2026-08-28 16:05:49	41.98	Locke	MIDDLE	FALSE	15	9	5	328	7.81	34138	19296	48	2.22
KR_8358902857	2026-08-28 15:26:20	28.68	Syndra	MIDDLE	FALSE	7	9	4	240	8.37	29744	12739	32	1.22
KR_8358865136	2026-08-28 15:03:19	15.30	Ryze	MIDDLE	TRUE	10	0	3	150	9.80	12462	9009	12	13.00
KR_8358744273	2026-08-28 14:12:44	31.55	Akali	MIDDLE	TRUE	15	5	10	274	8.68	50569	17931	30	5.00
KR_8357524539	2026-08-27 15:51:04	35.53	Zoe	MIDDLE	FALSE	8	10	10	223	6.28	29377	14257	33	1.80
KR_8357454848	2026-08-27 15:03:40	31.97	Camille	SUPPORT	FALSE	2	10	14	54	1.69	17884	10129	115	1.60
KR_8357402664	2026-08-27 14:34:51	21.28	Camille	TOP	TRUE	7	2	7	196	9.21	24819	11155	15	7.00
KR_8349819732	2026-08-21 17:10:07	24.37	Galio	MIDDLE	TRUE	16	1	11	230	9.44	36932	15106	23	27.00

match_id	game_time	duration_min	champion	position	win	kills	deaths	assists	cs	cs_per_min	damage	gold	vision_score	kda
KR_8349804117	2026-08-21 16:46:53	15.88	Tristana	MIDDLE	FALSE	1	2	3	138	8.69	13392	5954	13	2.00
KR_8349767265	2026-08-21 16:23:47	17.08	Syndra	MIDDLE	FALSE	6	4	4	125	7.32	16419	7219	15	2.50
KR_8349718982	2026-08-21 16:03:05	15.55	Syndra	MIDDLE	TRUE	13	2	6	142	9.13	23918	10484	10	9.50
KR_8349638350	2026-08-21 15:12:23	30.03	Locke	MIDDLE	FALSE	10	11	2	236	7.86	30114	15945	29	1.09
KR_8347432922	2026-08-19 16:14:13	29.65	TwistedFate	MIDDLE	TRUE	5	4	16	262	8.84	23367	15337	22	5.25
KR_8346438082	2026-08-18 18:15:44	27.58	Camille	TOP	FALSE	15	8	5	113	4.10	23541	11380	28	2.50
KR_8346404476	2026-08-18 17:37:12	25.40	Galio	MIDDLE	FALSE	5	7	10	201	7.91	18841	11084	33	2.14
KR_8346359120	2026-08-18 16:28:03	34.45	Orianna	MIDDLE	TRUE	9	10	16	242	7.02	43174	15812	28	2.50
KR_8346320115	2026-08-18 15:50:55	32.65	Smolder	MIDDLE	FALSE	9	8	6	286	8.76	48639	15092	55	1.88
KR_8345589624	2026-08-18 06:34:06	23.77	Syndra	MIDDLE	TRUE	18	3	12	222	9.34	43187	15583	25	10.00
KR_8340465799	2026-08-14 16:50:07	25.73	Cassiopeia	MIDDLE	FALSE	5	11	2	206	8.01	24193	9916	31	0.64
KR_8340442123	2026-08-14 16:25:50	15.92	Locke	MIDDLE	FALSE	2	6	2	123	7.73	10508	5835	8	0.67
KR_8339025982	2026-08-13 14:58:16	30.22	Akali	MIDDLE	FALSE	5	9	9	226	7.48	28049	12780	20	1.56
KR_8338008838	2026-08-12 17:12:53	24.90	Galio	MIDDLE	TRUE	7	3	10	186	7.47	18681	10835	25	5.67
KR_8337981651	2026-08-12 16:41:26	23.55	Locke	MIDDLE	FALSE	5	6	3	169	7.18	15543	8868	17	1.33
KR_8337952733	2026-08-12 16:10:26	22.08	Cassiopeia	MIDDLE	FALSE	3	6	3	176	7.97	19803	8453	27	1.00
KR_8337906728	2026-08-12 15:31:15	16.80	Akali	MIDDLE	TRUE	8	1	3	159	9.46	16586	8721	21	11.00
KR_8337787333	2026-08-12 14:10:37	22.10	Sylas	JUNGLE	FALSE	7	10	9	116	5.25	26905	9074	23	1.60
KR_8337700740	2026-08-12 13:29:57	23.50	Tristana	MIDDLE	TRUE	6	1	6	254	10.81	23474	12195	33	12.00
KR_8336831195	2026-08-11 17:07:17	24.60	Locke	MIDDLE	TRUE	11	6	7	196	7.97	26480	13088	26	3.00
KR_8336798143	2026-08-11 16:33:09	18.97	Cassiopeia	MIDDLE	TRUE	10	4	5	163	8.59	18567	8843	14	3.75
KR_8336741621	2026-08-11 15:50:01	15.17	LeeSin	JUNGLE	TRUE	7	1	10	118	7.78	10461	7725	23	17.00
KR_8336689125	2026-08-11 15:03:01	29.75	Ryze	MIDDLE	TRUE	10	6	12	227	7.63	32547	14187	29	3.67
KR_8335387019	2026-08-10 13:58:30	32.93	Anivia	MIDDLE	FALSE	7	7	13	244	7.41	28687	13797	43	2.86
KR_8335312980	2026-08-10 13:30:28	21.05	Locke	MIDDLE	TRUE	13	3	6	162	7.70	20707	11160	21	6.33
KR_8331199059	2026-08-07 15:47:05	21.02	Locke	MIDDLE	FALSE	8	9	1	130	6.19	18503	9451	16	1.00
KR_8331153208	2026-08-07 15:23:30	15.45	Cassiopeia	MIDDLE	TRUE	6	3	4	129	8.35	11360	7159	18	3.33

Player Performance Comparison

```r
library(ggplot2)
library(dplyr)
library(knitr)
library(kableExtra)


# ============================================================
# PLAYER PERFORMANCE COMPARISON TABLE
# ============================================================

player_comparison <- data.frame(

Player = c(
  "Solar",
  "Faker",
  "Chovy",
  "ShowMaker",
  "Zeka"
),


# -------------------------
# Combat Statistics
# -------------------------

Average_KDA = c(
  mean(solar_matches$kda, na.rm = TRUE),
  mean(faker_matches$kda, na.rm = TRUE),
  mean(chovy_matches$kda, na.rm = TRUE),
  mean(showmaker_matches$kda, na.rm = TRUE),
  mean(zeka_matches$kda, na.rm = TRUE)
),


Average_Kills = c(
  mean(solar_matches$kills, na.rm = TRUE),
  mean(faker_matches$kills, na.rm = TRUE),
  mean(chovy_matches$kills, na.rm = TRUE),
  mean(showmaker_matches$kills, na.rm = TRUE),
  mean(zeka_matches$kills, na.rm = TRUE)
),


Average_Deaths = c(
  mean(solar_matches$deaths, na.rm = TRUE),
  mean(faker_matches$deaths, na.rm = TRUE),
  mean(chovy_matches$deaths, na.rm = TRUE),
  mean(showmaker_matches$deaths, na.rm = TRUE),
  mean(zeka_matches$deaths, na.rm = TRUE)
),


Average_Assists = c(
  mean(solar_matches$assists, na.rm = TRUE),
  mean(faker_matches$assists, na.rm = TRUE),
  mean(chovy_matches$assists, na.rm = TRUE),
  mean(showmaker_matches$assists, na.rm = TRUE),
  mean(zeka_matches$assists, na.rm = TRUE)
),



# -------------------------
# Farming Statistics
# -------------------------

Average_CS = c(
  mean(solar_matches$cs, na.rm = TRUE),
  mean(faker_matches$cs, na.rm = TRUE),
```

```
  mean(chovy_matches$cs, na.rm = TRUE),
  mean(showmaker_matches$cs, na.rm = TRUE),
  mean(zeka_matches$cs, na.rm = TRUE)
),


Average_CS_Per_Min = c(
  mean(solar_matches$cs_per_min, na.rm = TRUE),
  mean(faker_matches$cs_per_min, na.rm = TRUE),
  mean(chovy_matches$cs_per_min, na.rm = TRUE),
  mean(showmaker_matches$cs_per_min, na.rm = TRUE),
  mean(zeka_matches$cs_per_min, na.rm = TRUE)
),



# -------------------------
# Economy / Map Impact
# -------------------------

Average_Vision_Score = c(
  mean(solar_matches$vision_score, na.rm = TRUE),
  mean(faker_matches$vision_score, na.rm = TRUE),
  mean(chovy_matches$vision_score, na.rm = TRUE),
  mean(showmaker_matches$vision_score, na.rm = TRUE),
  mean(zeka_matches$vision_score, na.rm = TRUE)
),


Average_Damage = c(
  mean(solar_matches$damage, na.rm = TRUE),
  mean(faker_matches$damage, na.rm = TRUE),
  mean(chovy_matches$damage, na.rm = TRUE),
  mean(showmaker_matches$damage, na.rm = TRUE),
  mean(zeka_matches$damage, na.rm = TRUE)
),


Average_Gold = c(
  mean(solar_matches$gold, na.rm = TRUE),
  mean(faker_matches$gold, na.rm = TRUE),
  mean(chovy_matches$gold, na.rm = TRUE),
  mean(showmaker_matches$gold, na.rm = TRUE),
  mean(zeka_matches$gold, na.rm = TRUE)
),



# -------------------------
# Game Context
# -------------------------

Average_Game_Duration = c(
  mean(solar_matches$duration_min, na.rm = TRUE),
  mean(faker_matches$duration_min, na.rm = TRUE),
  mean(chovy_matches$duration_min, na.rm = TRUE),
  mean(showmaker_matches$duration_min, na.rm = TRUE),
  mean(zeka_matches$duration_min, na.rm = TRUE)
)

)



# Round values

player_comparison <- player_comparison %>%
  mutate(
    across(
      where(is.numeric),
```

```
      round,
      2
    )
  )
```

```
## Warning: There was 1 warning in `mutate()`.
## ℹ In argument: `across(where(is.numeric), round, 2)`.
## Caused by warning:
## ! The `...` argument of `across()` is deprecated as of dplyr 1.1.0.
## Supply arguments directly to `.fns` through an anonymous function instead.
##
##   # Previously
##   across(a:b, mean, na.rm = TRUE)
##
##   # Now
##   across(a:b, \(x) mean(x, na.rm = TRUE))
```

```
# Display table

kable(
  player_comparison,
  caption = "Average Performance Statistics — Last 50 Ranked Games",
  row.names = FALSE
) %>%
kable_styling(
  font_size = 9,
  full_width = TRUE
)
```

```
## Warning: 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```

```
## Warning: 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```

Average Performance Statistics — Last 50 Ranked Games

Player	Average_KDA	Average_Kills	Average_Deaths	Average_Assists	Average_CS	Average_CS_Per_Min	Average_Vision_Score	Average_Damage	Average_Gold	Average_Game_Durat
Solar	1.77	7.56	10.04	7.46	141.28	4.95	22.78	22851.86	11513.48	28
Faker	4.71	5.86	4.90	6.82	206.28	8.04	33.36	22786.08	11433.82	26
Chovy	4.49	5.92	4.08	6.88	230.72	9.13	41.00	24310.00	12030.38	25
ShowMaker	4.00	6.26	4.60	6.60	199.28	7.81	25.98	23079.96	11422.98	25
Zeka	4.70	7.72	5.66	7.24	187.04	7.73	29.20	24242.36	11399.64	24

```
# ============================================================
# PLAYER PERFORMANCE RANKING GRAPHS
# ============================================================

library(ggplot2)
library(dplyr)


# ============================================================
# CREATE RANKING PLOT FUNCTION
# ============================================================

create_ranking_plot <- function(data, metric, title, decreasing = TRUE) {

  # Arrange ranking order
  if (decreasing) {
    plot_data <- data %>%
      arrange(desc(.data[[metric]]))
  } else {
    plot_data <- data %>%
      arrange(.data[[metric]])
  }

  # Fix player order for ggplot
  plot_data$Player <- factor(
    plot_data$Player,
    levels = rev(plot_data$Player)
  )


  ggplot(
    plot_data,
    aes(
      x = .data[[metric]],
      y = Player
    )
  ) +

    # Line from zero to value
    geom_segment(
      aes(
        x = 0,
        xend = .data[[metric]],
        y = Player,
        yend = Player
      ),
      linewidth = 0.5
    ) +

    # Ranking point
    geom_point(
      size = 4
    ) +

    # Value label
    geom_text(
      aes(
        label = .data[[metric]]
      ),
      hjust = -0.25,
      size = 4
    ) +

    # Prevent labels being cut off
    scale_x_continuous(
      expand = expansion(
        mult = c(0.05, 0.20)
      )
    ) +
```

```
    labs(
      title = title,
      x = "",
      y = ""
    ) +

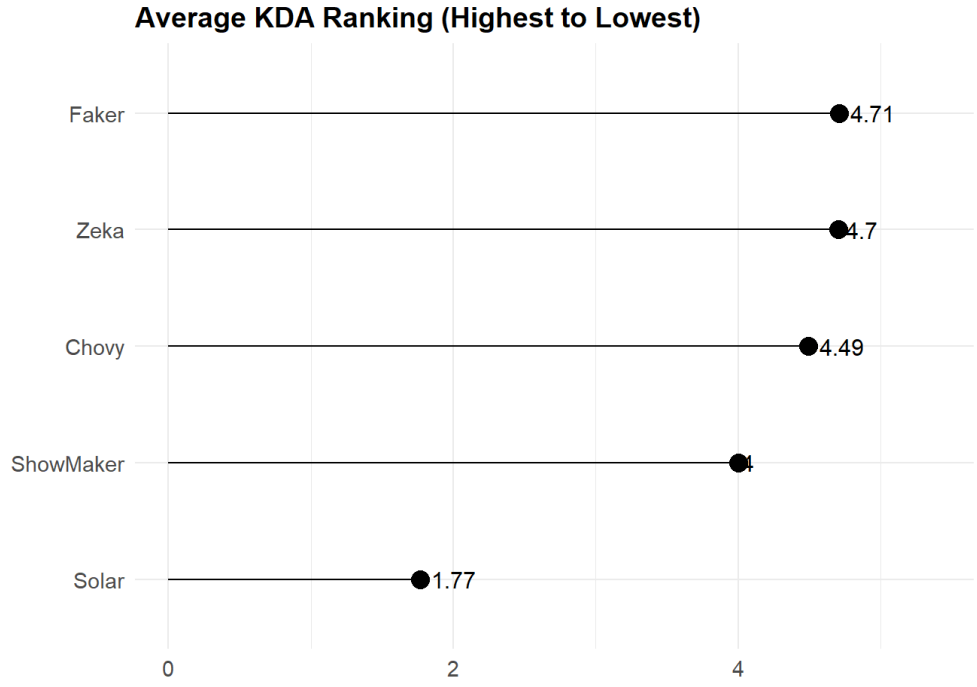
    theme_minimal() +

    theme(
      plot.title = element_text(
        size = 14,
        face = "bold"
      ),
      axis.text = element_text(
        size = 11
      )
    )
}



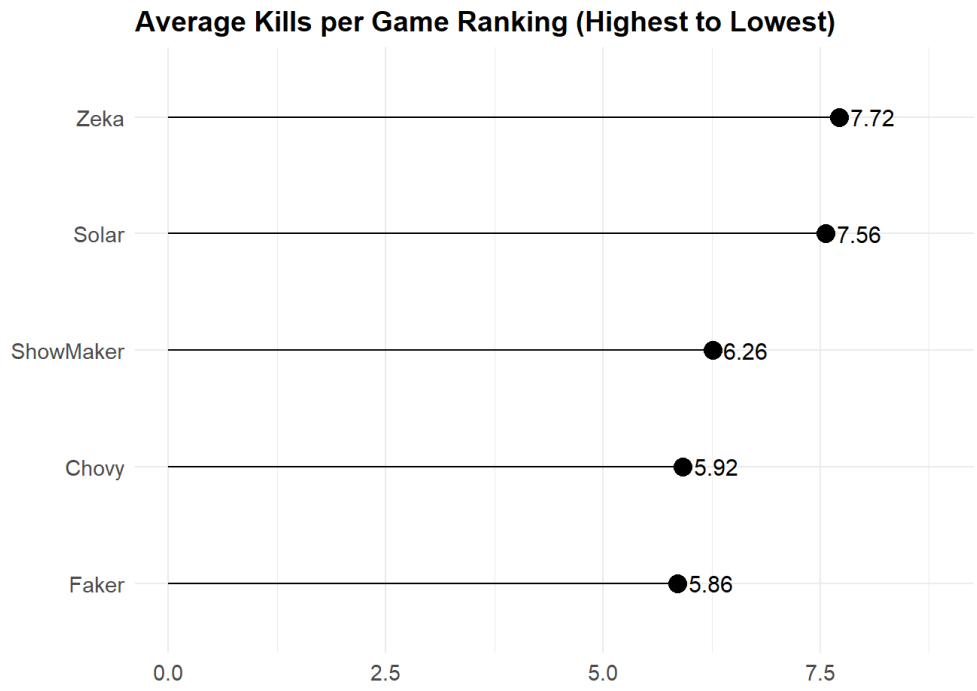
# ============================================================
# KDA
# ============================================================

create_ranking_plot(
  player_comparison,
  "Average_KDA",
  "Average KDA Ranking (Highest to Lowest)",
  TRUE
)
```



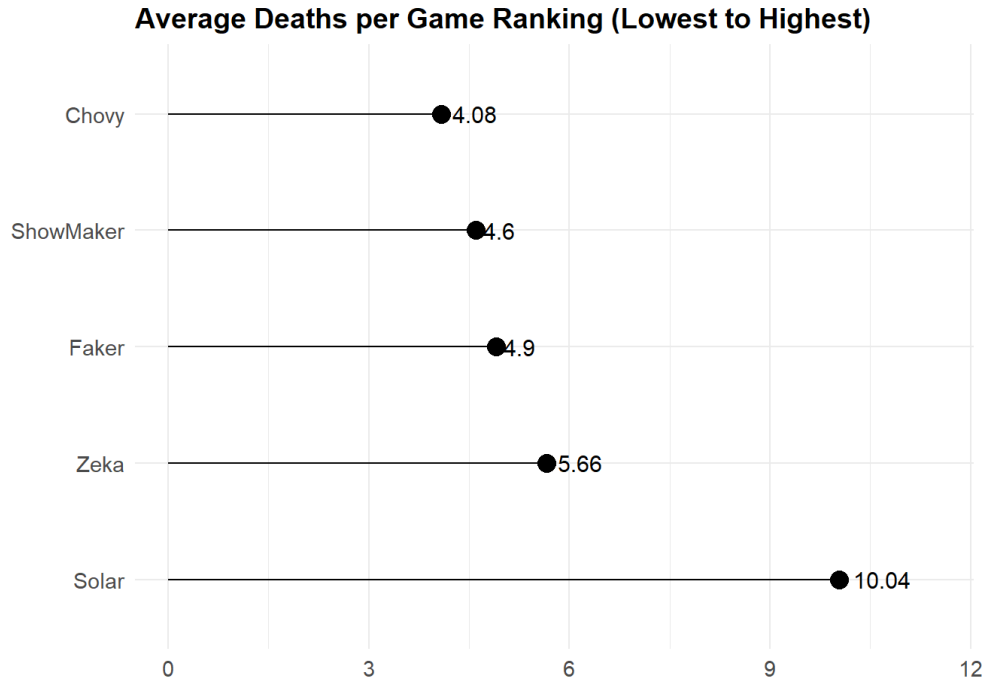
```
# ============================================================
# KILLS
# ============================================================

create_ranking_plot(
  player_comparison,
  "Average_Kills",
  "Average Kills per Game Ranking (Highest to Lowest)",
  TRUE
)
```



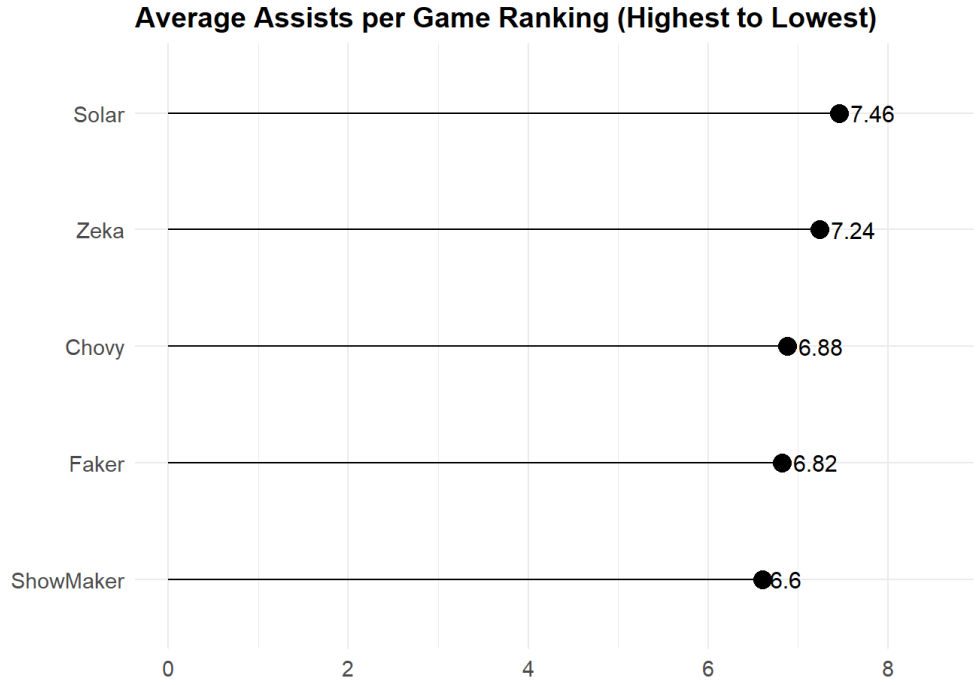

```
# ============================================================
# DEATHS
# Lowest is better
# ============================================================

create_ranking_plot(
  player_comparison,
  "Average_Deaths",
  "Average Deaths per Game Ranking (Lowest to Highest)",
  FALSE
)
```



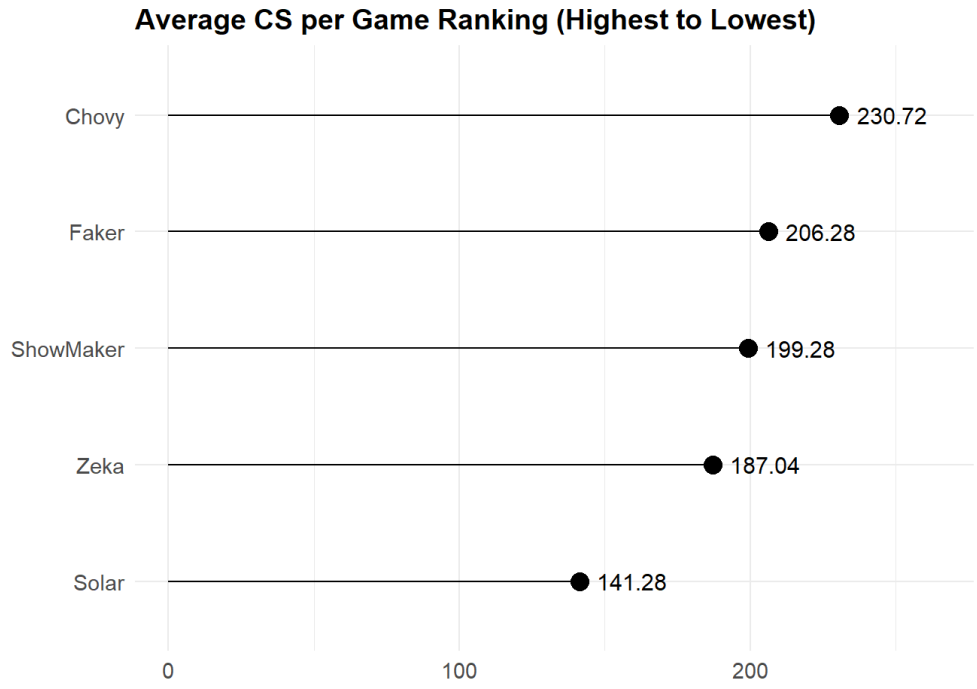

```
# ============================================================
# ASSISTS
# ============================================================

create_ranking_plot(
  player_comparison,
  "Average_Assists",
  "Average Assists per Game Ranking (Highest to Lowest)",
  TRUE
)
```



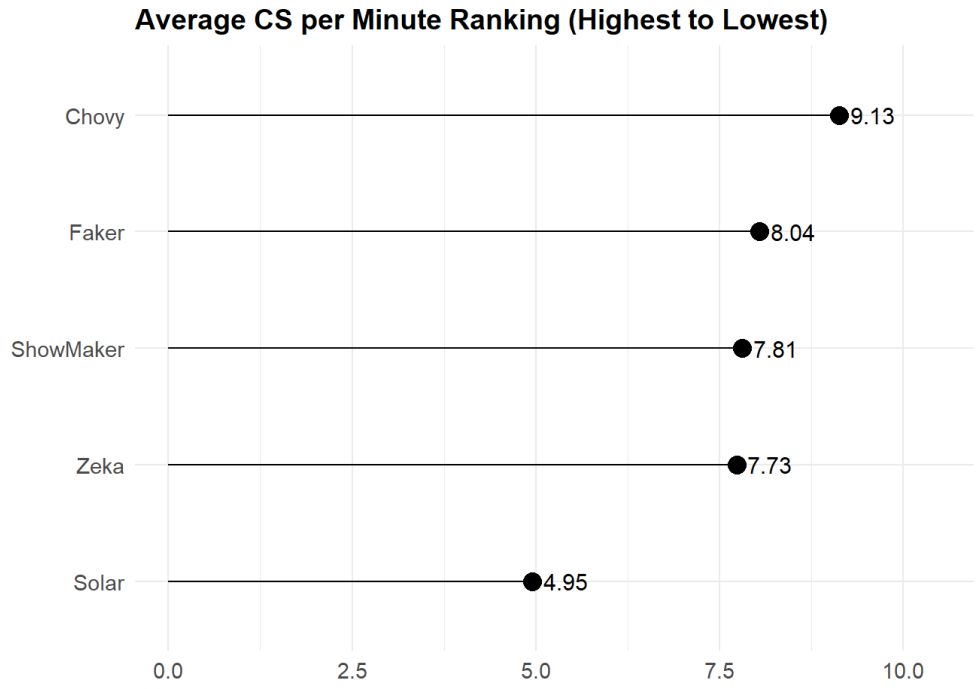
```
# ============================================================
# CS
# ============================================================

create_ranking_plot(
  player_comparison,
  "Average_CS",
  "Average CS per Game Ranking (Highest to Lowest)",
  TRUE
)
```



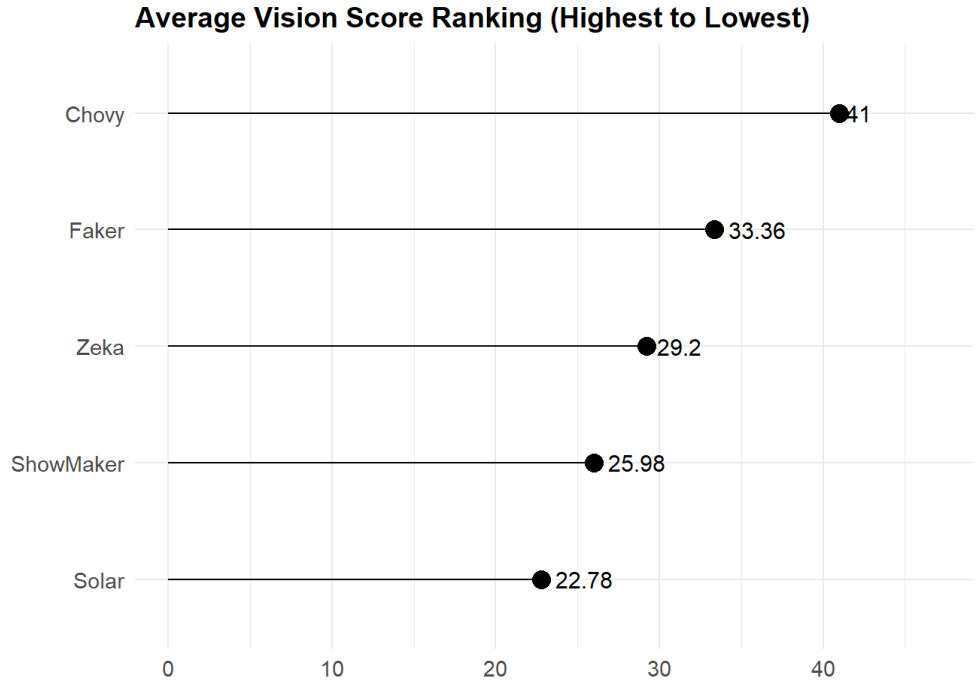

```
# ============================================================
# CS PER MINUTE
# ============================================================

create_ranking_plot(
  player_comparison,
  "Average_CS_Per_Min",
  "Average CS per Minute Ranking (Highest to Lowest)",
  TRUE
)
```



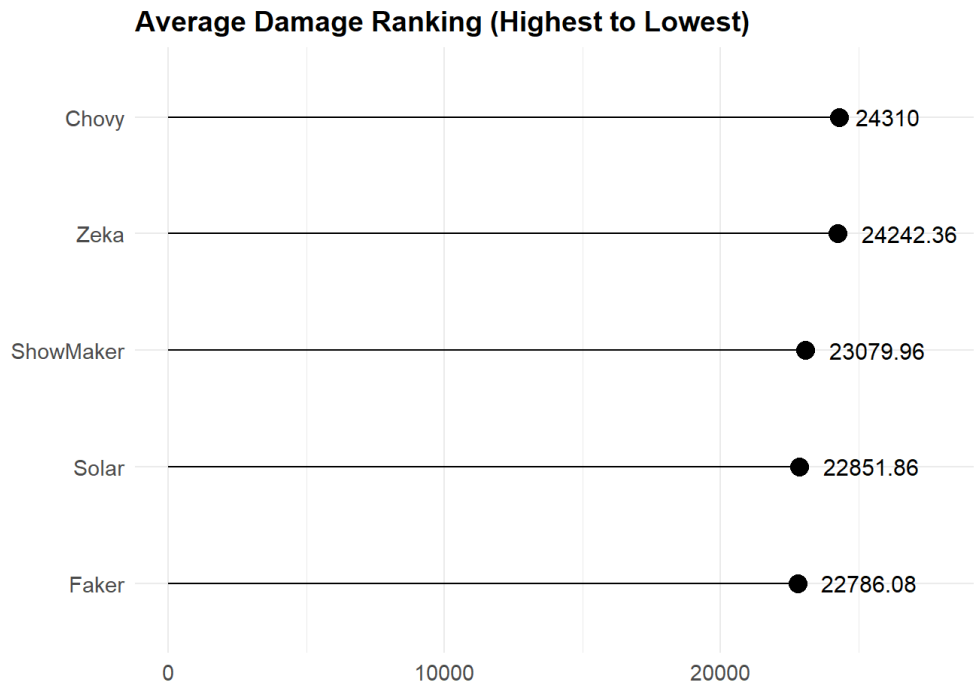

```
# ============================================================
# VISION SCORE
# ============================================================

create_ranking_plot(
  player_comparison,
  "Average_Vision_Score",
  "Average Vision Score Ranking (Highest to Lowest)",
  TRUE
)
```



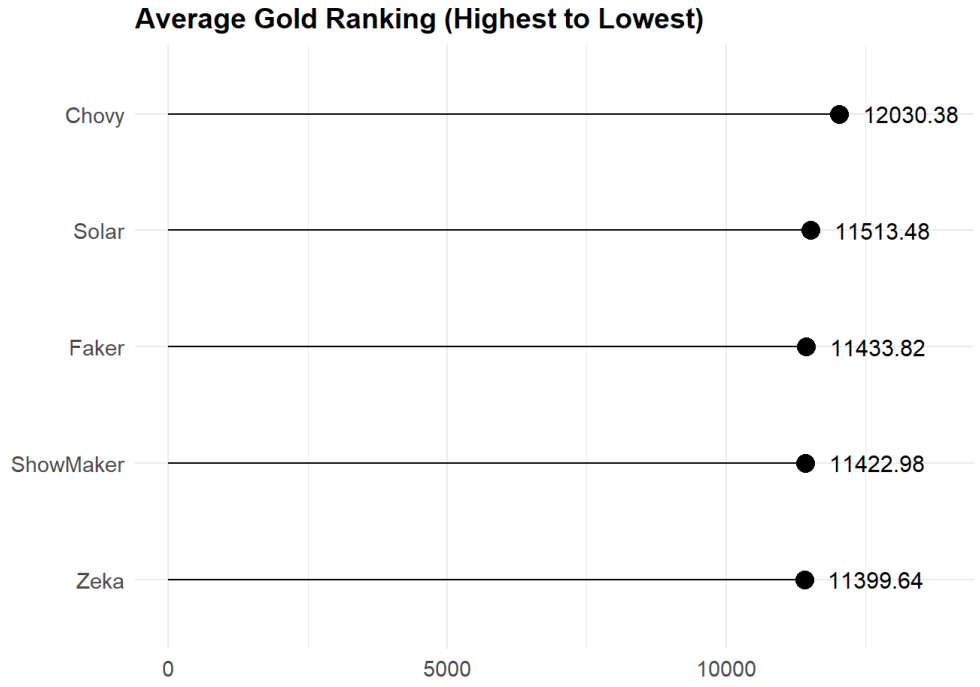
```
# ============================================================
# AVERAGE DAMAGE
# ============================================================

create_ranking_plot(
  player_comparison,
  "Average_Damage",
  "Average Damage Ranking (Highest to Lowest)",
  TRUE
)
```




```
# ============================================================
# AVERAGE GOLD
# ============================================================

create_ranking_plot(
  player_comparison,
  "Average_Gold",
  "Average Gold Ranking (Highest to Lowest)",
  TRUE
)
```




```
# ============================================================
# GAME DURATION
# ============================================================

create_ranking_plot(
  player_comparison,
  "Average_Game_Duration",
  "Average Game Duration Ranking (Highest to Lowest)",
  TRUE
)
```

